Greecon
SH.P.K

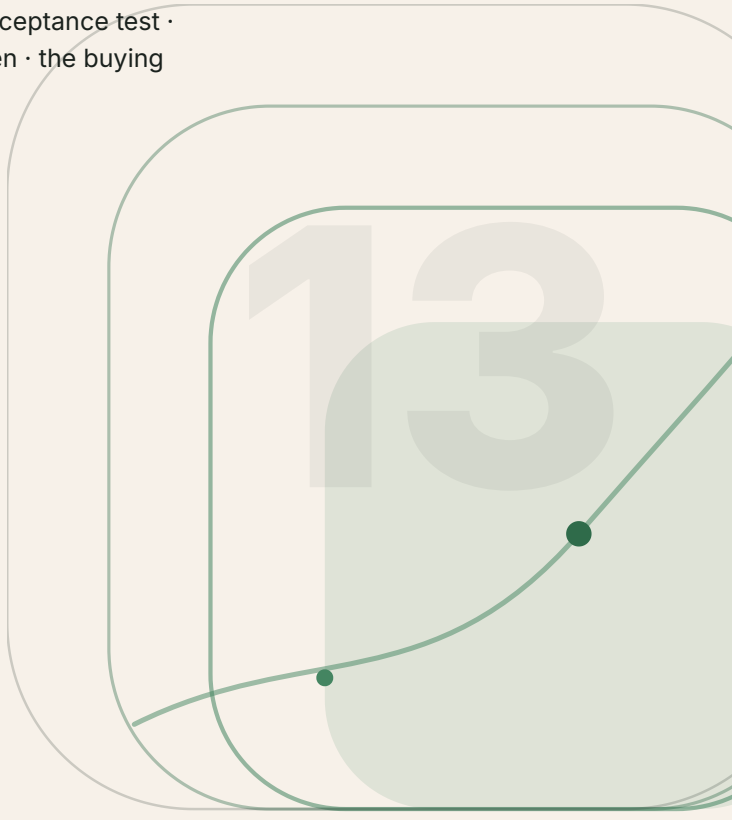
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HARDWARE SHEET HW13 · QUALIFIED BY ITS REGISTER TABLE

Soil moisture probe

The probe in the soil of each irrigation zone. No model is approved yet, so its address and its register are entered by hand.

QUALIFIES	Qualified by its register table. A soil probe that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
USED IN	Kit F (H hotel or guesthouse, S solar plant, F farm or greenhouse, W water tank and pump station)
FOR	The project manager at a farm or a greenhouse
COVERS	How it qualifies · what it does at each kind of site · settings · fitting · in the platform · the acceptance test · what goes wrong · what is not yet proven · the buying record



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HW13

Soil moisture probe

A soil moisture probe sits in the soil of an irrigation zone and says how wet the soil is at its depth. It speaks Modbus RTU on the same data cable as everything else. **No model of soil probe is approved yet.** Greecon does not know any probe's registers, and the box cannot recognise one on a network. So a probe qualifies by its maker's register table, its address is entered by hand, and its reading is mapped by hand from that table.

At a glance

WHAT IT DOES	Measures soil moisture at the probe's depth. Its depth and place are written down, because soil readings mean little without them. Unit 6, mapped by hand from the maker's register table.
HOW IT QUALIFIES	**Qualified by its register table.** A soil probe that speaks Modbus RTU on RS-485, whose maker publishes its register table, and whose number and speed can be set.
SPECIFICATION	RS-485 Modbus, with a published register table, one for each zone
NUMBER ON THE CABLE	6 (a second probe takes the next free number, 8). Set the maker's way, at the office, before the probe is buried.
FITTED BY	The project manager, with the grower, who says where and how deep.
PLANNING ESTIMATE (EUR)	40 to 120 each. Not a quote.
USED IN	Kit F. The table under *At each kind of site* says what it does at each.

How it qualifies

It qualifies if:

- ☐ It speaks **Modbus RTU on RS-485,** on two data wires.
- ☐ **Its maker publishes its register table,** and you have it in your hand before you buy: the register that holds the moisture, the kind of number it is, and its scale.
- ☐ Its unit number and its speed can be set, and the maker says how: to a number of its own, and to 9600 baud, 8 data bits, no parity, 1 stop bit.
- ☐ What it reports becomes a percentage. The reading in Greecon is **Soil moisture**, from 0 to 100, and the table says how the probe's number becomes one.
- ☐ It is made to stay in the soil, and its cable reaches a dry junction with no joint in the ground.
- ☐ Its supply voltage is one the kit has: read its label against the 24 V supply.

STOP

Do not buy a probe whose register table you have not seen. A probe whose maker does not document its registers cannot be mapped honestly, so it cannot be used. **And never bury a probe before its number is set:** one that was not must be dug up.

At each kind of site

Kind of site	In its kit	How many	What it does there
H Hotel or guesthouse	NOT USED	None	None.
S Solar plant with grid connection	NOT USED	None	None.
F Farm or greenhouse with irrigation	IN THE KIT	1 for each zone	One for each irrigation zone, at the depth and in the place agreed with the grower. It gives Soil moisture. Greecon gives no irrigation advice until the grower's own Start irrigating below value is entered.
W Water tank and pump station	NOT USED	None	None.

What Greecon reads from it

One reading for each probe: **Soil moisture**, in percent.

What is real	0 to 100. Outside that, Greecon says that the number cannot be soil moisture
Standard alert	None
The limit	*Start irrigating below*. It has **no standard value**: the same percentage is waterlogged in sand and bone dry in clay, and it depends on the crop, the depth of the roots and the probe
Until the limit is entered	Greecon gives no irrigation advice. It shows the reading and its trend
Once it is entered	Greecon says when the soil has fallen below it. That is ordinary work on a farm, not an alert

Soil readings are only as good as the contact between the probe and the soil, and they mean little without the probe's depth and place. Both are written down.

Settings, and where they are made

Setting	Value	Where
Unit number	**6.** A second probe, for a second zone, takes the next free number: 8, then 9. Many probes are delivered as 1, the same as another device. Set it the maker's way, with the USB-to-RS-485 adapter and no other device on the cable	At the office
Speed	9600 baud, 8 data bits, no parity, 1 stop bit, set the maker's way	At the office
Its label	The number, written on the probe's cable at the end that stays above the ground	At the office

Fitting

PROJECT MANAGER with the grower, who says where and how deep.

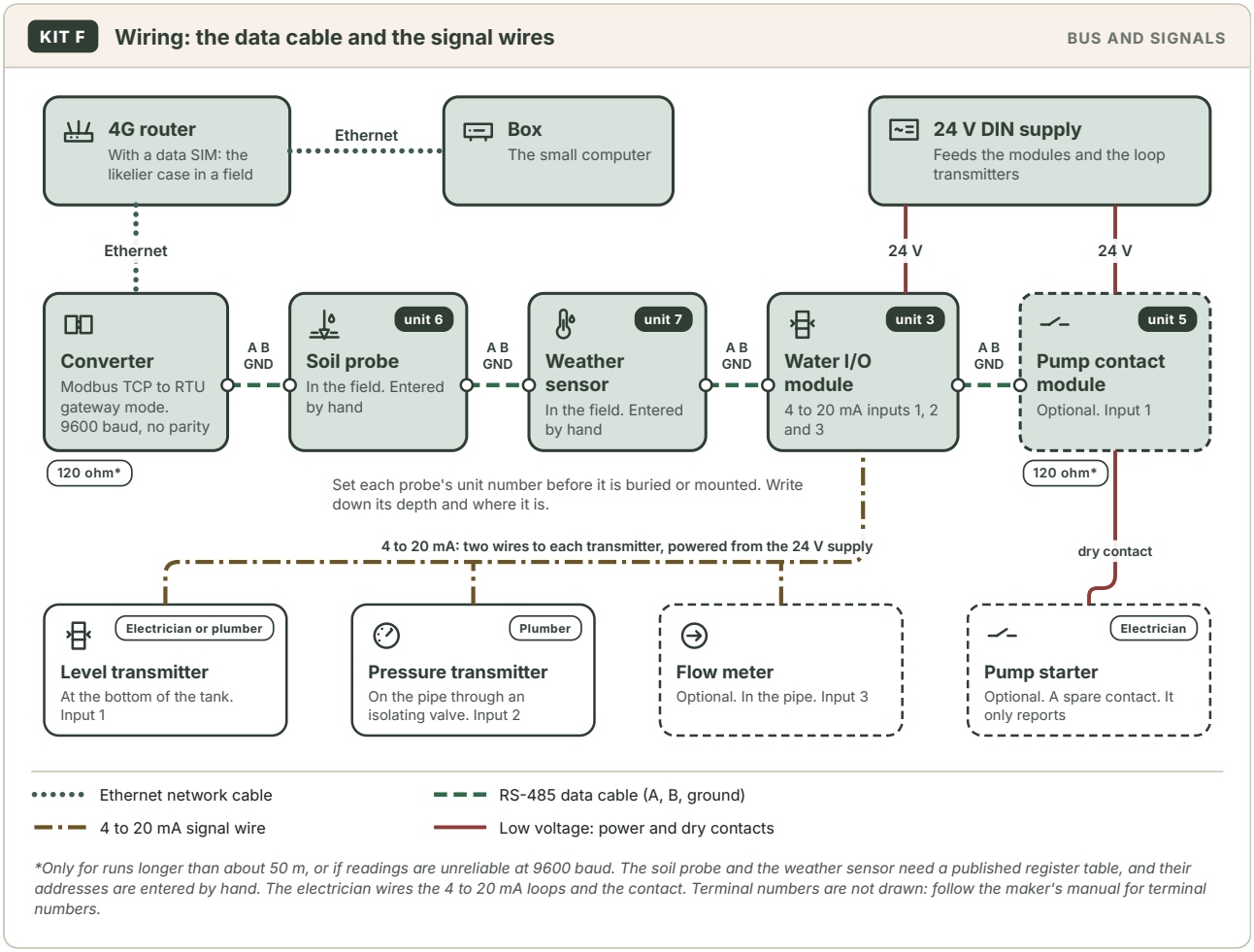


Figure F-2. Kit F. Wiring: one RS-485 cable from the converter through the soil probe, the weather sensor and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module.

☐ **The place is typical of the zone:** not its wettest spot and not its driest. Walk the field with the grower and choose it together.

- [ ] **The depth is where the roots are.** Ask the grower.
- [ ] The probe is placed as its maker says, **in good contact with the soil,** with no air gap around it.
- [ ] It joins the data cable in the daisy chain: A to A, B to B, no branches (sheet HW03). Plan the whole run under 100 metres.
- [ ] **Write down each probe's depth and place, and photograph it** before the soil is closed.

In the platform

The farm template makes the device **Soil probe** with its reading and no register.

1. **Its address** (step 4, **Where each device is on the network**). The box cannot recognise a probe, so its line says **Enter by hand**. Under the device open **Enter the address by hand** and fill in the **IP address** (the converter's: the water module's line shows it, as *Reached at 192.168.1.50:502, unit 3*), the **Port** (502) and the **Unit number** (6). Press **Save**.
2. **Its register** (step 5, **Which register holds which reading**). Under the reading press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx. The address counts from zero (30001 is address 0). Choose the kind of number, and under how it becomes the reading choose **Multiply and add**: a probe that reports tenths of a percent is multiplied by 0.1. Press **Save the register**.
3. **A second zone.** On the site's **Equipment and readings** page, under *Add equipment or a reading by hand (advanced)*, add a second soil probe and its reading. Then give it its address and its register as for the first, with its own unit number (8).
4. **The limit** (step 10, **Alert limits agreed with the customer**). Leave *Start irrigating below* empty on the day. After the first week of readings, ask the grower which reading means "time to water", and enter it.

Write what you enter:

Probe	Unit number	Register kind	Address	Kind of number	Multiply by	Depth and place
Zone 1	6					
Zone 2	8					

Acceptance test

At the office, on the desk, before anything is buried:

- [ ] The probe has its number (6), alone on the cable, and is labelled.
- [ ] Behind the converter, its register is read from the box, and the number is believable:

```
greecon-edge modbus-test --host 192.168.1.50 --unit 6 --type input --address 0 --data-type uint16
--scale 0.1
```

(The kind, the address, the kind of number and the scale are the maker's: these are an example.)

- [ ] In the air it reads low, and in a glass of water it reads high.

On the day, as part of commissioning:

Check and how	Pass when	Greecon shows	Real	Result
Soil probe. In the air, then in a glass of water, then in the soil.	Three different readings: low in the air, high in the water, plausible in the soil			☐ Pass ☐ Fail

What goes wrong

What you see	Likely cause	What to do
The probe does not answer	A wrong unit number (left as delivered, the same as another device), a number or an address entered wrongly, a speed that is not the cable's, A and B swapped, or no power	Check each, in that order. Test the register from the box against the maker's table
The reading never changes, or is the same in the air and in water	A wrong register, the wrong register kind (Input against Holding), or a wrong scale	Read the maker's table again and test the register from the box
The reading is ten times too big or too small	The scale	Under **Multiply and add**, multiply by the table's factor
The reading is believable but does not follow watering	Poor contact with the soil, or a place that is not typical of the zone	Place the probe again, with the grower. Write the new depth and place
No irrigation advice is shown	*Start irrigating below* has no standard value	After the first week, ask the grower and enter it (step 10)

More is in the runbooks (Project Manager Field Manual, PDF 02): R2 (a device does not answer) and R3 (a reading looks wrong).

Not yet proven

- **No soil probe has been read by this software, and no model is approved.** Every make is a first time: the test above is the proof, for that probe.
- Greecon says nothing about what a soil reading means for a crop. The limit is the grower's own, and Greecon never waters anything: it only reads.
- The prices are planning figures, not quotes.

Buying record

Ask three suppliers for a quote and keep the quotes in the shared drive. Buy one, pass the acceptance test with it, and only then buy more. For every part that is bought the hardware register keeps the make and model, the supplier, the date, the serial number or the batch, and the invoice.

	Quote 1	Quote 2	Quote 3
Supplier			
Make and model offered			
Price (EUR, without VAT), and delivery			
Meets every line of *It qualifies if*	☐ Yes	☐ Yes	☐ Yes

Bought	Answer
Make, model and version	
Supplier, date and invoice	
Serial number or batch	
Acceptance test passed on (date), by	